Peter Sitati

Technology Leader · AI, Fintech & Platform Engineering

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SUMMARY

Innovative technology leader with 15+ years building scalable software platforms, leading cross-functional teams, and driving digital transformation across fintech, media, agritech, and infrastructure. Proven track record shipping production AI systems including voice-driven applications, multi-channel agentic platforms, LLM-powered automation, financial reconciliation and data-migration pipelines, alongside mobile apps, payment systems, and enterprise integrations. Starts by asking what the brief left out, and writes up every change so the next engineer can see why.

PROFESSIONAL EXPERIENCE

Chief Executive Officer · **NARS Technical Services** *Mar 2022 to Present · Nairobi, Kenya*

- Designed and shipped a production **LLM-as-judge evaluation system** (DSPy) for multi-tier grading of open-ended learner answers with rubric dimensions, confidence scoring and human-review escalation, inside a **voice-first AI learning platform** (LiveKit/WebRTC, LangGraph, Deepgram, ElevenLabs) with provider failover across OpenAI/Google/Soniox, subscription billing, and 2,300+ automated tests on Fly.io, with grading and speech-to-text accuracy ratcheted into CI.
- Architected a **multi-channel AI lead-nurturing engine** (LangGraph + FastAPI) coordinating voice (VAPI), WhatsApp, and SMS with Claude conversational logic, a Qdrant **RAG** knowledge base, and **M-Pesa**-integrated multi-tenant billing, serving multiple enterprise clients.
- Developed a **settlement reconciliation platform** (FastAPI, React, LangGraph) automating extraction of bank settlement data across **38 African countries**: NIBSS, KEPSS, PAPSS, SWIFT.
- Engineered an **enterprise data-migration pipeline** ingesting ~12,000 legacy financial documents into ERPNext via a 4-stage parser chain with LLM fallback (Claude/Cerebras) and vector-embedding semantic deduplication (ChromaDB).
- Created an **AI content management agent** (Mastra, Claude) to autonomously grade, optimize SEO/GEO, and curate **1,200+ blog posts** for an EV publication focused on developing economies.
- Built **Run.KE**, a running events platform for Kenya (Wasp 0.21, React, Node.js, Prisma): event discovery and admin tooling with **LLM-vision flyer parsing** for stylized posters, running-track and gym directories (OpenStreetMap), a route tool, Cloudflare R2 presigned uploads, Turnstile bot protection, and scheduled analytics jobs.
- Built an **agentic content manager** (Mastra, Claude) for a health publication that grades drafts across mission-alignment and technical-quality dimensions, optimizes SEO, and publishes to Ghost CMS under YMYL/E-E-A-T standards.
- Integrated **gbrain**, a local-first **knowledge-graph memory system** served over **MCP**, across my most important repositories as a working **knowledge base**, writing the ingestion, rescan, remote-sync and health-check tooling around it and the query playbooks it is used through: semantic retrieval over local Ollama embeddings in PGLite, graph traversal, entity/fact extraction, and contradiction detection over the corpus.
- Built a **tender document automation tool** (Python, PyMuPDF, Claude API) and **eTIMS integration tooling** for KRA tax-invoice compliance (20+ endpoints); engineered a DeFi arbitrage trading bot (Solidity, Web3.py).
- Led deployment of secure, high-availability networking infrastructure (fiber, wireless, LAN) for enterprise clients.

Head of Mobile · **TradePulse Inc.** *Jan 2024 to Aug 2025 · Nairobi, Kenya*

- Led end-to-end Flutter development of the **TradePulse Mobile App**, enabling agro-processors to manage inventory, create orders, accept payments, and generate reports.
- Spearheaded React/Next.js development of the **TradePulse Sourcing Platform**, a CRM-driven B2B system streamlining raw-material sourcing between buyers and suppliers; scaled regionally to hundreds of daily transactions.

Team Leader, Innovation / Software Engineer · **Radio Africa Group** *May 2016 to Apr 2022 · Nairobi, Kenya*

- Directed agile teams building consumer digital products reaching **millions of users**: a music streaming platform with real-time recommendations, a digital newspaper with subscriptions, and interactive games/quizzes.
- Championed modern software practices and cloud infrastructure to accelerate development cycles and digital transformation.

Software Engineer · **Cellulant Kenya** *Nov 2011 to Apr 2016 · Nairobi, Kenya*

- Developed and maintained **bulk payment systems** for enterprise and government clients, enabling high-volume secure transactions.
- Built payment systems powering the **Growth Enhancement Support Program** (Nigeria), connecting input suppliers to **millions of farmers** to deliver subsidies efficiently.

Bioinformatician · IITA · Earlier roles *2005 to 2011*

- Built data-mining and bioinformatics workflows for genomic data analysis at IITA (2010 to 2011). Prior: Research Assistant, Innovative Health Research · Network Installations Coordinator, NARS · Drupal Integrator, Koumbit Network (Montreal) · Web Developer, Norpix Inc. & Canadian Chemical Biology Network.

SELECTED PROJECTS

- **Quizlar**: Voice-first AI learning platform with real-time WebRTC quiz sessions and FSRS spaced repetition, exposed to external agents through an MCP server. LiveKit · LangGraph · DSPy · Deepgram · FastAPI · Flutter · React · PostgreSQL
- **Multi-Channel Voice Agent Engine**: Agentic voice/WhatsApp/SMS lead-nurturing platform with RAG and M-Pesa billing. VAPI · Claude · Qdrant · MySQL + PostgreSQL · React/TypeScript
- **TradePulse Sourcing Platform**, B2B supply-chain CRM for agricultural sourcing: role-based dashboards for administrators, agro-processors and suppliers, covering aggregated demand, supplier records, quote lifecycle, deals, orders, payments and wallets. Next.js 15 · React · TypeScript
- **Doc Sift**: Document extraction pipeline turning invoices, POs, receipts and bank statements into structured records synced to the client's system of record, with an eval harness on hand-checked ground truth. TypeScript · LLM extraction
- **Settlement Reconciliation Platform**: 15-stage pipeline normalising ten settlement formats (M-PESA, MTN, Airtel, NIBSS, KEPSS, PAPSS, SWIFT MT940/MT103, ISO 20022 camt.053) across 38 African countries. Python · FastAPI · React · LangGraph
- **ERPNext Accounting Ingestion**: 12k-document financial migration with LLM-fallback parsing and vector dedup. Python · ChromaDB · Docling · Claude/Cerebras · ERPNext REST API
- **Run.KE**, Running events platform for Kenya: event discovery/admin with LLM-vision flyer parsing, track & gym directories, route tool. Wasp 0.21 · React/TypeScript · Node.js · Prisma · Cloudflare R2 + Turnstile
- **Healthspan Content Agent**: Agentic content manager that grades drafts on mission-alignment and technical quality, optimizes SEO, and publishes to Ghost CMS. Mastra · Claude API · Ghost CMS · TypeScript
- **Cross-repository knowledge base**: gbrain, a local-first MCP knowledge-graph memory system, integrated across my most important repositories, with the ingestion, rescan, sync and health-check tooling written around it. gbrain · PGLite · Ollama embeddings · MCP · Claude API
- **Plopped**, Bristol-scale bowel health diary, shipped solo to iOS: Premium subscription (in-app purchase), engagement-gated ads, hands-free voice logging (on-device STT into LLM field extraction, reviewed before save), biometric lock, timezone-aware reminders, offline-first local store. Documented domain model and ADRs; CI-built releases. Flutter · Riverpod · Drift (SQLite) · Firebase Analytics · AdMob · StoreKit

TECHNICAL SKILLS

Languages & Frameworks: Flutter, FlutterFlow, React, Next.js, Node.js, Wasp (Open SaaS), FastAPI, JavaScript, TypeScript, Python, PHP, Solidity, SQL, Bash

AI & Voice: LLM Evaluation & Evals (LLM-as-judge), LangGraph, DSPy, Mastra, n8n, Claude API, OpenAI, MCP (built a FastMCP server), Qdrant (RAG), ChromaDB, PGLite, Ollama, Deepgram (STT), ElevenLabs (TTS), LiveKit (WebRTC), VAPI

Data & Platforms: PostgreSQL, MySQL, SQLite, Redis, AWS, GCP, Fly.io, Docker, n8n, Ghost CMS, CI/CD, REST APIs

Other: LLM Evaluation Design, AI Agent Orchestration, RAG Systems, Data Extraction Pipelines, Systems Architecture, M-Pesa Integration, Agile Leadership

EDUCATION

Jomo Kenyatta University of Agriculture & Technology
BSc Biochemistry & Molecular Biology

McGill University
Biochemistry & Molecular Biology coursework, credits transferred to JKUAT

MORE

Languages: Swahili (Native), English (Native), French (Working)

Certification: Java Developer Certificate

Publication: *Genomics & Potential Downstream Applications in the Developing World*